



# Koneru Lakshmaiah Education Foundation

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## OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

### Project Problem Statement Submission Form

#### Section / Team Details

Section No.: 1

Team No.: 21

**Project Title:** Linux System Information and Resource Monitoring Tool

#### Team Members

| Roll Number | Student Name      | Signature |
|-------------|-------------------|-----------|
| 2520030088  | J.Sai Satyan      |           |
| 2520030084  | CH.Vishnu Vardhan |           |
| 2520030523  | B.Sandeep         |           |
|             |                   |           |
|             |                   |           |

#### 1. Abstract

Modern operating systems manage a wide range of hardware and software resources such as the CPU, memory, disk, and processes, often without giving users a clear, real-time view of system health. On Linux systems such as Ubuntu, this information is scattered across virtual filesystem interfaces like `/proc` and `/sys`, and command-line utilities like `top`, `free`, and `df`, making it inconvenient to monitor resource usage holistically. This project proposes the design and implementation of a Linux System Information and Resource Monitoring Tool, a lightweight utility that consolidates key system metrics, CPU utilization, memory usage, disk statistics, running processes, and system uptime, into a single, readable interface.

The tool directly applies core Operating Systems and Systems Programming concepts, including process management, virtual memory, file systems, and system calls. It reads live data from the `/proc` filesystem (e.g., `/proc/stat`, `/proc/meminfo`, `/proc/[pid]/status`) using Linux system calls and POSIX APIs, avoiding reliance on third-party monitoring software. The proposed methodology involves periodically polling these kernel-exposed data sources, parsing the raw output, and presenting it through a structured terminal-based dashboard. The expected outcome is a functional, low-overhead monitoring utility that accurately reflects real-time system resource usage on Ubuntu, demonstrating practical understanding of how the Linux kernel exposes system state to user-space programs.

#### 2. Problem Statement

Linux systems expose a wealth of information about their internal state, but this information is fragmented across many separate interfaces (e.g., `/proc`, `/sys`) and disparate command-line tools, with no single lightweight utility that consolidates it. Users, students, and system administrators often cannot easily assess CPU load, memory pressure, disk usage, and process activity together, forcing repeated context switching between multiple commands. This problem is directly relevant to Operating Systems and Systems Programming, as solving it requires understanding how the Linux

kernel exposes process, memory, and file-system state to user space, and how system calls and pseudo-file interfaces can be used programmatically to build a monitoring utility.

### 3. Objectives

1. To design and implement a Linux-based tool that collects real-time CPU, memory, disk, and process information using kernel-exposed interfaces (/proc, /sys).
2. To display the collected system metrics in a clear, organized, and continuously updating command-line interface.
3. To apply core Operating Systems concepts such as process management, memory management, and file systems in a practical programming context.
4. To evaluate the tool's accuracy and performance overhead against standard Linux utilities such as top, free, and df.

### 4. Proposed Methodology

The tool will be developed in C/C++ on Ubuntu Linux, using the /proc and /sys virtual filesystems as the primary data sources. A core data-collection module will periodically read files such as /proc/stat, /proc/meminfo, /proc/diskstats, and /proc/[pid]/status, parsing their contents to extract structured metrics for CPU usage, memory consumption, disk activity, and running processes.

This data will be aggregated and refreshed at fixed time intervals (e.g., every 1-2 seconds) and rendered through a text-based dashboard in the terminal. The design will follow a modular architecture separating data collection, parsing, and display logic, allowing individual components to be tested, extended, or replaced (e.g., with a GUI or logging module) without affecting the rest of the system.

### 5. Operating Systems Concepts / Linux APIs Used

| OS Concept / Linux API / System Call   | Purpose in the Project                                                                   |
|----------------------------------------|------------------------------------------------------------------------------------------|
| Process Management (/proc/[pid])       | Retrieve process states, PIDs, and resource usage for each running process               |
| Virtual Memory (/proc/meminfo)         | Read total, used, free, and cached memory statistics                                     |
| File Systems (VFS, /proc, /sys)        | Access kernel-exposed pseudo-files to read live system state without direct kernel calls |
| System Calls (open, read, close)       | Perform low-level file I/O to read data from /proc and /sys efficiently                  |
| CPU Scheduling Statistics (/proc/stat) | Compute CPU utilization by sampling jiffies counters over time                           |
| Signal Handling (SIGINT)               | Gracefully handle termination signals to clean up resources on exit                      |

### 6. Individual Contribution

| Roll Number | Student Name | Individual Responsibility |
|-------------|--------------|---------------------------|
| 2520030088  | J.Sai Satyan | Abstract , ppt ,UI        |
| 2520030084  | ChVishnu     | Coding, CLI, abstract     |
| 2520030523  | B.Sandeep    | Testing and GUI           |
|             |              |                           |
|             |              |                           |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                                                                            |
|----------------------------|------------------------------------------------------------------------------------|
| Linux / Ubuntu             | Target operating system and development/testing environment                        |
| C / C++                    | Primary programming language for accessing system calls and /proc data efficiently |
| GCC / Make                 | Compiling and building the project                                                 |
| ncurses (optional)         | Building an interactive terminal-based dashboard UI                                |
| Git / GitHub               | Version control and project collaboration                                          |

## 8. Expected Outcome

The project is expected to produce a functional command-line utility that displays real-time CPU utilization, memory usage, disk statistics, and a list of active processes on an Ubuntu Linux system, refreshed at regular intervals.

The tool will demonstrate practical application of Operating Systems concepts and provide a lightweight, dependency-free alternative to combining multiple standard Linux monitoring commands, with potential for future extension into logging, alerting, or a graphical interface.

## 9. GitHub Repository

**GitHub Repository Link:** [https://github.com/chvishnuvardhanchowdary2007/Espadas\\_OSSP](https://github.com/chvishnuvardhanchowdary2007/Espadas_OSSP)

**Repository Created:** \_\_\_\_\_

☐ Yes   ☐ No

## Faculty Name and Signature

**Faculty Name:** \_\_\_\_\_

**Signature:** \_\_\_\_\_

**Remarks:** \_\_\_\_\_

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## Faculty Use

**Project Approved:** \_\_\_\_\_

☐ Yes   ☐ No

**Date:** \_\_\_\_\_

**Faculty Signature:** \_\_\_\_\_